

# Odari Tutorials

Made Easy

## Control Systems Engineering Tutorial

### Time-Domain Analysis of a Second-Order System

#### 1. Core Problem Statement

A closed-loop position control system is characterized by the following second-order transfer function:

$$T(s) = \frac{C(s)}{R(s)} = \frac{25}{s^2 + 6s + 25}$$

Assuming the system is initially at rest and is subjected to a unit-step input,  $R(s) = \frac{1}{s}$ , determine:

The system parameters:

- Un-damped natural frequency ( $\omega_n$ )
- damping ratio ( $\zeta$ )
- damped natural frequency ( $\omega_d$ ).

The definitive analytical time-response expression,  $c(t)$ .

The transient response performance specifications ( $t_r$ ,  $t_p$ ,  $M_p$ , and  $t_s$  using a 2% criterion).

#### 2. Step-by-Step Analytical Solution

##### Part A: Extraction of System Parameters

The characteristic equation of a standard second-order system is defined as:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

By equating the coefficients of the standard form to the system's denominator ( $s^2 + 6s + 25 = 0$ ):

**Natural Frequency ( $\omega_n$ ):**

$$\omega_n^2 = 25 \Rightarrow \omega_n = 5 \text{ rad/s}$$

**Damping Ratio ( $\zeta$ ):**

$$2\zeta\omega_n = 6 \Rightarrow 2\zeta(5) = 6 \Rightarrow \zeta = 0.6$$

*Classification:* Because  $0 < \zeta < 1$ , the system is **fundamentally underdamped**.

**Damped Natural Frequency ( $\omega_d$ ):**

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} = 5\sqrt{1 - 0.6^2} = 5(0.8) \\ \Rightarrow \omega_d = 4 \text{ rad/s}$$

*Part B: Derivation of the Time Response  $c(t)$*

The general time-domain solution for an underdamped unit-step response is given by the formula:

$$c(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \theta)$$

First, solve for the phase angle ( $\theta$ ) in radians:

$$\theta = \tan^{-1} \left( \frac{\sqrt{1-\zeta^2}}{\zeta} \right) = \tan^{-1} \left( \frac{0.8}{0.6} \right) \approx \mathbf{0.927} \text{ rad} \quad (53.13^\circ)$$

Substituting the calculated system constants yields the final time-response equation:

$$\mathbf{c(t) = 1 - 1.25e^{-3t} \sin(4t + 0.927)}$$

*Part C: Calculation of Transient Specifications*

Rise Time ( $t_r$ ): The time required for the response to first reach 100% of its final value.

$$t_r = \frac{\pi - \theta}{\omega_d} = \frac{\pi - 0.927}{4} \approx \mathbf{0.554} \text{ seconds}$$

Peak Time ( $t_p$ ): The time required to reach the maximum peak of the response.

$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{4} \approx \mathbf{0.785} \text{ seconds}$$

Maximum Percent Overshoot ( $M_p$ ): The maximum peak value measured from the final steady-state value.

$$M_p = e^{-\left(\frac{\zeta\pi}{\sqrt{1-\zeta^2}}\right)} \times 100\% = e^{-\left(\frac{0.6\pi}{0.8}\right)} \times 100\% \approx \mathbf{9.48\%}$$

Settling Time ( $t_s$ ): The time required for the response curve to settle within a  $\pm 2\%$  tolerance band of the final value.

$$t_s = \frac{4}{\zeta\omega_n} = \frac{4}{3} \approx \mathbf{1.333} \text{ seconds}$$

### 3. Verification Baseline Reference Table

This structured dataset serves as your direct point of comparison when verifying your results using MATLAB or Python plots in the video tutorial.

| Performance Metric | Mathematical Symbol | Analytical Exact Value |
|--------------------|---------------------|------------------------|
| Natural Frequency  | $\omega_n$          | 5.000 rad/s            |
| Damping Ratio      | $\zeta$             | 0.600                  |
| Damped Frequency   | $\omega_d$          | 4.000 rad/s            |
| Rise Time          | $t_r$               | 0.554 s                |
| Peak Time          | $t_p$               | 0.785 s                |
| Maximum Overshoot  | $M_p$               | 9.48%                  |
| Settling Time (2%) | $t_s$               | 1.333 s                |

